



A deep sign language recognition system for Indian sign language

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Abstract

Deaf people face major challenges during communication with normal people. Employing a human interpreter (a person who converts Sign language (SL) into a language that the normal/hearing community can understand) is not an effective solution to this problem due to the unavailability of professional interpreters. Thus, the Sign Language Recognition System (SLRS) is the most efficient and effective choice because it automatically converts SL into text/speech without an interpreter and reduces the communication barrier between deaf and normal people. This paper reports a work on Indian Sign Language (ISL) word recognition using a vision-based technique. The existing vision-based solutions for ISL word recognition are ineffective due to excessive pre-processing such as extracting features from a large sequence of frames. Therefore, a vision-based SLRS named Hybrid CNN-BiLSTM SLR (HCBSLR) is proposed, which overcomes the drawback of excessive pre-processing. The proposed model uses a Histogram Difference (HD) based key-frame extraction method to improve the accuracy and efficiency of the system by eliminating redundant or useless frames. The HCBSLR system uses VGG-19 for spatial feature extraction and Bidirectional Long Short Term Memory (BiLSTM) for temporal feature extraction. The proposed HCBSLR system has achieved an average accuracy of 87.67%, which is compared with some of the existing SLRS. The experimental results show that the proposed HCBSLR system is more accurate and efficient than the existing SLRS.

Keywords Machine learning · Deep learning · Key-frame · Sign language

1 Introduction

For the deaf community, Sign Language (SL) is an important and alternative mode of communication, and Sign Language Recognition Systems (SLRS) are advanced technology to aid the communication process. In the realm of human–computer interaction, automatic SL recognition has become one of the most pressing issues. The goal of SLRS is to automatically translate signs into text or speech. SLRS is not only vital to bridge the communication gap between deaf and hearing individuals but also to expand

the quantity of content available to the deaf, such as the production of deaf community-friendly educational tools or games and visual dictionaries of SL. In past studies, wearable sensors such as data gloves, accelerometers, or vision-based systems have been proposed as a solution to Sign Language Recognition (SLR) problem. The intrinsically noisy and ambiguous signs as input data, is an issue for vision-based systems, regardless the type of inputs whether these are RGB images or depth information. On the other hand, data gloves produce more accurate and descriptive results. However, vision-based SLR systems are more preferred for real-world applications because they are less complicated and do not require bulky gadgets while performing any sign. Over the past ten years, several vision based SLR models have been proposed, with improved recognition performance. The introduction of deep learning methodologies, notably Convolutional Neural Networks (CNN), has aided a significant impact of this recent success.

There are two types of SLRS: Static SLRS and Dynamic SLRS. Dynamic SLRS are classified into isolated and

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continuous SLRS. The static SLRS recognizes SLs that do not have any motion associated with them, while the dynamic SLRS recognizes SLs with motion. Many Machine Learning (ML) and Deep Learning (DL) based solutions are proposed by researchers to recognize SL. In case of ML based solution, the selection of relevant feature descriptors for detecting SL is a lengthy trial and error procedure [1]. Furthermore, traditional machine learning-based SLR systems necessitate a significant amount of pre-processing, making this approach inefficient. Traditional ML-based models have performed effectively for sign alphabets recognition, but not for sign words recognition, which incorporates dynamic gestures. Deep learning-based solutions are capable of resolving this problem. Deep learning-based CNN can be used as an automatic feature extractor to avoid the necessity of manual feature extraction. The use of CNN makes the SLR system more efficient, as this can avoid the pre-processing steps. CNN requires a large dataset for training a model; however, unavailability of large dataset and limited hardware resources are the major concerns for training a CNN model. Therefore, to resolve this issue, a pre-trained CNN model is used that utilizes the concept of transfer learning, which reduces the requirement of large training dataset and training time.

This work focuses on Indian Sign Language (ISL) recognition which is used in India as an alternative to spoken language. The majority of ISL recognition research is focussed on static SL recognition or alphabet recognition [2–7]. However, in real-world applications, advanced SLRS is needed to recognize words or sentences that integrate motion gestures. Dynamic SL recognition is substantially more challenging than static SL recognition due to its longer sequences and co-relation among multiple gestures. Many solutions are proposed to resolve the dynamic SL recognition problem, but most of them are sensor-based approaches [8, 9]. A handful of studies have employed a vision-based approach, but they are limited to a smaller vocabulary [10–14]. Sridhar et al. [15] have developed a large vocabulary of Indian Sign Language (ISL) and proposed a vision-based ISL recognition system where they have used an open pose model for the extraction of three sets of features: key points, pose video and Part Affinity Fields (PAF) video. They have used different combination of features with different models and encoder–decoder model with pose video feature has been found the best where CNN is used as an encoder and bidirectional long short-term memory (BiLSTM) as a decoder. Aparna et al. [16] have used a CNN-LSTM architecture for ISL recognition task where InceptionV3 is used as pre-trained CNN model for extracting spatial features from a sequence of frames and a stacked long short-term memory (LSTM) is used as a sequence learner. The dataset proposed by [15]

has resolved the unavailability of publicly available ISL dataset issue; however, the open pose model has decreased the efficiency of the SLRS as extracting pose or PAF feature from a large sequence of frames requires a lot of time and computational power. SLR model proposed by [16] has used a simple LSTM network as a sequence learner. Simple LSTM network gives output based on previous input and present input, whereas BiLSTM network utilizes two LSTM layers: one layer in the forward direction and another layer in the backward direction; as a result, the final output depends not only on the present and past input, it also depends upon future data that give potentially better result compared to simple LSTM network for SL recognition task. Hence, both the models are inefficient for SL recognitions efficiently and effectively.

To overcome the problems cited above, this research article proposes a model termed as Hybrid CNN-BiLSTM SLR (HCBSLR) which attempts to encounter the issue of extracting features from a large sequence of frames with the help of key-frame extraction algorithm. The proposed HCBSLR model extracts key-frame using histogram difference (HD) technique from a large sequence of frames efficiently and increases the efficiency and recognition accuracy by removing redundant and useless frames. The proposed work utilizes a CNN-BiLSTM architecture where different pre-trained CNN models are tested as a feature extractor and BiLSTM network as a sequence learner and the combination of VGG-19 and BiLSTM has been found the best for SL recognition system. This combination is used for the first time in case of SL recognition task. The proposed HCBSLR model has reached an average accuracy of 87.67% for the selected twenty commonly used ISL words. The obtained result from the proposed HCBSLR model seems promising for dynamic ISL recognition. The reported work will benefit the deaf community since it prepares a path for developing a communication platform.

The remainder of the paper is structured as follows: Literature survey of SLRS is presented in Sect. 2. Section 3 gives detail of the proposed method for ISL word recognition. Experimental details are given in Sect. 4 and the experimental evaluation of the proposed methodology is presented in Sect. 5, which includes a comparison with state-of-the-art methodologies. Section 6 concludes with a summary of the findings as well as some suggestions for future research.

2 Literature survey

SLR system identifies SL gestures and translates them into a text or speech that are meaningful to the hearing community. The SL vocabulary is a collection of ordered gestures so the SLR systems are closely linked with the

human gesture recognition task or human action recognition task. Over the years, researchers have proposed different models for sign language recognition as well as human gesture recognition. These models are divided into two categories based on their approaches: hardware-based and vision-based. Hardware-based models have a lower level of complexity than vision-based models. However, they have several significant downsides, such as signers face difficulties while performing any gesture, and this method is not cost-effective, which makes this approach incompatible with real-world applications. To solve this, a new vision-based technique that is better suited to SL recognition is employed. Researchers have developed various SLR systems over the last decade, using traditional ML and advanced DL methodologies. This section discusses current advances in SL recognition using ML and DL algorithms and their drawbacks and benefits.

Traditional machine learning algorithms classify a sequence of frames or images that reflect a specific sign word or sign gesture by extracting spatial and temporal information. Traditional techniques such as image pre-processing, image segmentation, hand detection, contour detection, hand shape detection, and hand tracking are utilized as required steps, followed by feature extraction and classification. Some of the ISL recognition systems have worked in this direction. Dutta et al. [17] have used principal component analysis (PCA) as feature extractor and K-nearest neighbour (KNN) for classification. The proposed method has got an accuracy of 95.84% for ISL alphabet recognition. Unlike other SL, ISL consists of single- as well as double-handed sign gesture. For single- and double-handed SL recognition, Gupta et al. [18] have used a fusion-based technique where histogram-oriented gradient (HOG) and scale invariant feature transformation (SIFT) are used with KNN for recognizing ISL alphabets. The proposed method has achieved an accuracy of 90%. Most of the ISL recognition systems based on machine learning techniques are designed for static gesture recognition or ISL alphabets recognition as extracting frames from a SL video that consist dynamic gesture and then applying image processing techniques to extract hand region requires a lot of time which makes SLR system inefficient. To resolve the issue, Athira et al. [19] have proposed a key-frame extraction-based SLR system for ISL word recognition which consist of dynamic gesture, where Zernike moments (ZM) is used for key-frame extraction which reduces the number of unwanted frames significantly. The proposed method has reached an accuracy of 89% for ISL word with dynamic gesture. Choudhury et al. [20] have developed another method for recognition of continuous ISL that is based on conditional random field (CRF) framework. CRF framework has been used for segmenting and labelling sequential frames. The proposed method has reached an accuracy of 86% for double-handed

signs and 90% for single-handed sign. The performance of the machine learning-based SLR systems completely depends upon extracted features; so identification of accurate features with different shape in various background and lighting condition is a challenging task. Raheja et al. [21] have proposed a Hu moment-based SLR system for recognizing ISL words, where Hu moment is basically a location, shape and angle invariant moment. The proposed system has tested on selected four words and it has achieved an accuracy of 97.5%. Sruthi et al. [22] have developed an SLR system for ISL alphabets recognition. They have proposed a CNN model named as signet, which can recognize ISL alphabets based on supervised learning technique. The model has trained with 5157 images where they have used image augmentation to increase the training samples. The model has achieved an accuracy of 98.64%. Training a model from scratch requires lot of data, to encounter this problem, Jaswal et al. [23] have used a pre-trained VGG-16 network for recognition of double-handed ISL digits. In this work, the VGG-16 network is fine-tuned and trained on the double-handed ISL dataset. The proposed approach has achieved 97.21% accuracy. CNN models are good at extracting spatial features, but they fall short when it comes to extracting temporal features. Likhar et al. [24] have proposed an SLR system that recognizes continuous SL. The CNN-LSTM architecture is used in this method. They have tested the proposed model on 10 dynamic gestures, and it has achieved the f1 score of 99.20%. Jaydeep et al. [25] have proposed another SLRS that can recognize dynamic ISL words related to banking sector. They have used pre-trained InceptionV3 model for feature extraction and LSTM network for gesture recognition. The model has achieved an accuracy of 85%. The simple LSTM network has several drawbacks. Thus, many researchers have turned to modified LSTM networks for SL recognition. Mittal et al. [26] have proposed an SLR system for continuous SL recognition where a modified LSTM is used. The SLR system has achieved an accuracy of 72.3% for ISL sentence recognition and 89.5% for isolated ISL word recognition. Aly et al. [27] have proposed another SLR system where they have used a BiLSTM network for dynamic SL recognition. They have also compared the proposed model with LSTM. They have shown that the BiLSTM network has performed better compared to normal LSTM for the SL recognition task.

3 Methodology of the proposed HCBSLR system

This section discusses the architecture of proposed HCBSLR system that can recognize ISL words with motion gestures. The proposed HCBSLR system has utilized the concept of CNN-LSTM hybrid model. CNN-

LSTM model is popular for dynamic sign word recognition. Many SLR systems are proposed based on the CNN-LSTM model, in which the model is trained from scratch. However, training a model from scratch requires a large amount of data, which is the primary concern in this experiment. Therefore, a pre-trained CNN network is used to overcome this problem. In this proposed work, a pre-trained 2-D CNN model VGG-19 is used for extracting the spatial features from the key-frames generated from the original frame sequence. Dynamic SL recognition not just depends upon spatial features it extensively depends on temporal feature. Thus, a BiLSTM network is employed to get the temporal pattern of dynamic sign words from the feature vector sequence generated by VGG-19 network. This section covers the key components of the proposed HCBSLR system in depth. The key components of the proposed HCBSLR system are (i) Key-frame extraction, (ii) Spatial feature extraction, and (iii) Temporal feature extraction, which are discussed in detail. Figure 1 shows the overview of the HCBSLR system.

3.1 Dataset description

The unavailability large ISL dataset is a significant setback for developing an ISL recognition system. Most researchers have created their own datasets for developing ISL recognition systems due to the lack of publicly available dataset. The proposed work uses a data set developed by Sridhar et al. [15] and it is a publicly available large dataset. The dataset contains ISL words with dynamic gesture. The dataset has 4287 videos in 263 classes, which belong to 15 different categories such as “adjective”, “animal”, “clothes”, “colour”, “people”, “place”. The SL videos are performed by a group of people, including males and females with almost similar backgrounds, and the length of each video is variable. The average number of videos per class is 16.3, and the average length of the video is 2.57 s. The maximum and minimum lengths of the videos are 6.16 s and 1.28 s, respectively. The resolution of each video is 1920×1080 .

Due to limited hardware resources and to make the experiment simpler, a subset of the original data set is used in this experiment. Twenty frequently used essential words are selected from the original dataset that belongs to different categories and these are “Alright”, “Bed”, “Bedroom”, “Bird”, “Cool”, “Doctor”, “Door”, “Fish”, “Good”, “Healthy”, “Hot”, “Old”, “Patient”, “Sick”, “Student”, “Teacher”, “Tomorrow”, “Warm”, “Window”, “Yesterday”. For each class or sign word, the number of videos is different. Table 1 gives detail distribution of the data set used for this experiment. Figures 2 and 3 show the sample frame sequences of sign videos taken from the data set.

The original dataset contains 0.27 million frames which is a significant number that requires a lot of computation power and time for evaluation. Therefore, a smaller subset that contains 330 videos and 13,200 frames is used in this experiment.

3.2 Key-frame extraction

Key-frame extraction is the key concept that makes HCBSLR system more efficient. The dynamic sign word videos consist of many frames. Processing all the frames need a high computation power system, and it also takes a long time to process all the frames. The aim is to improve the efficiency of the SLR system for deployment on portable devices. In case of dynamic SL, the sign word consists of multiple gestures, and there are transition phases between two consecutive gestures. During this transition phase, we get many frames with unnecessary information other than key pose or gesture. Therefore, removing these frames improves the efficiency and accuracy of the system.

The basic idea behind this approach is to select some specific frames from a large sequence of frames which is significantly different in terms of gesture position, in other words the aim is to eliminate the redundant and transition frames from a large sequence of frames. The proposed work uses a modified HD algorithm for key-frame extraction which is based on absolute difference of histogram (ADH) algorithm proposed by Sheena et al. [28]. The algorithm executes in two phases: in first phase, the histogram is computed for each frame then the difference between two consecutive frames is calculated using Euclidian distance method. Several distance metrics including Euclidian distance, City Block distance, Manhattan distance and absolute distance methods are used for key-frame extraction, of which the Euclidian distance method has given the best result. After distance calculation, the mean and standard deviation is calculated from the series of distances. In second step, the threshold value is compared with distances, and based on that, key-frames are extracted.

Figure 4 provides an example of extracted key-frame from a large sequence of frames and Fig. 5 gives a comparative analysis between extracted number of frames and number of key-frames for each sign word.

The threshold value (Th) is calculated from mean and standard deviation. Here, μ denotes the mean and σ denotes the standard deviation. The formula is given as follows:

$$\text{Th} = \mu + \sigma$$

The algorithm for finding Key-frame is given below.

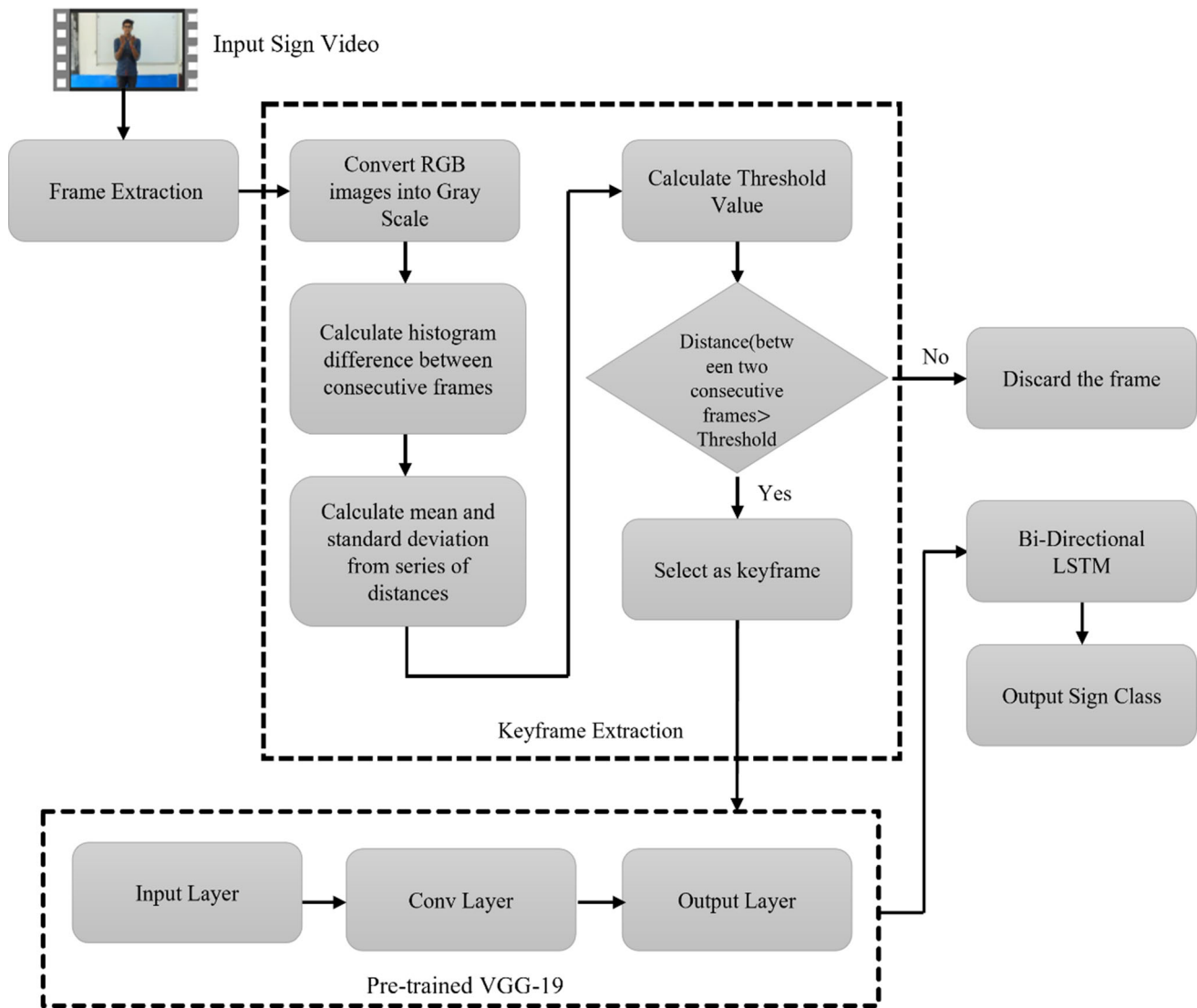


Fig. 1 Overview of proposed HCBSLR system

- Step1: Extract the frames from sign videos.
- Step2: Convert the RGB images into Gray scale.
- Step3: Calculate the mean and standard deviation of the histogram's Euclidean distance between consecutive frames.
- Step4: Calculate the threshold value.
- Step5: If Euclidian distance value $> Th$ then select the frame as Key-frame.
- Step6: Repeat the process for entire video.

3.3 Spatial feature extraction using pre-trained VGG-19 network

CNN is a deep learning model that is inspired by the operations of the human visual cortex for processing data with patterns, such as digital images and trained to

automatically learn spatial hierarchies of information from low level to high level. Convolution, pooling, and fully connected layers are the three types of layers or building blocks which are found in CNNs. The convolution and pooling layers aid in feature extraction, while the fully connected layer transfers those features into the final output and gives predictions based on probability. The activation function is an integral part of CNN, which is significant because they provide nonlinearity to the input data, which aids in capturing semantic diversity in input images by generating different patterns. The activation function and subsampling operations make the input data invariant of any geometrical changes [29, 30].

In this proposed work, the VGG-19 pre-trained CNN model is employed for spatial feature extraction from keyframes. The VGG-19 network has more depth than traditional convolutional neural networks. It switches between

Table 1 Class distribution of data set used for proposed SLRS

Name of the class	Performed by no. of people	Number of videos
Alright	6	18
Bed	7	21
Bedroom	4	12
Bird	6	18
Cool	7	21
Doctor	4	12
Door	4	12
Fish	6	18
Good	7	21
Healthy	7	21
Hot	7	21
Old	7	21
Patient	4	12
Sick	7	21
Student	4	12
Teacher	4	12
Tomorrow	4	12
Warm	7	21
Window	4	12
Yesterday	4	12
Total number of Frames	13,200	
Maximum duration of Sign video	2.04 s	
Minimum duration of Sign video	1.63 s	
Resolution	1920 × 1080	

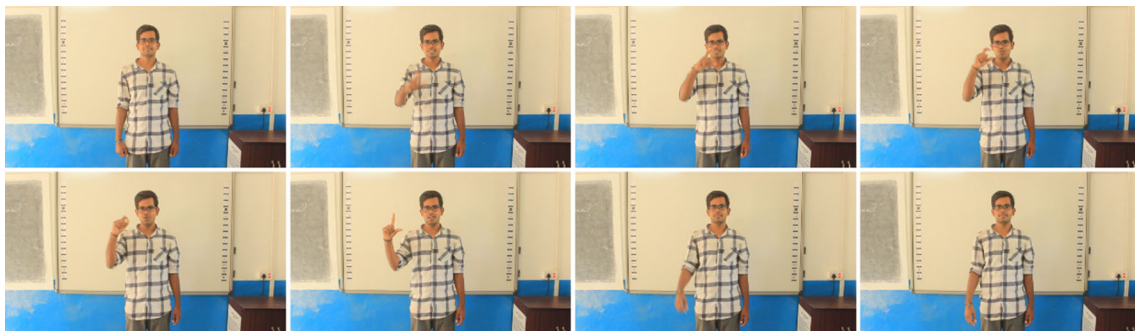
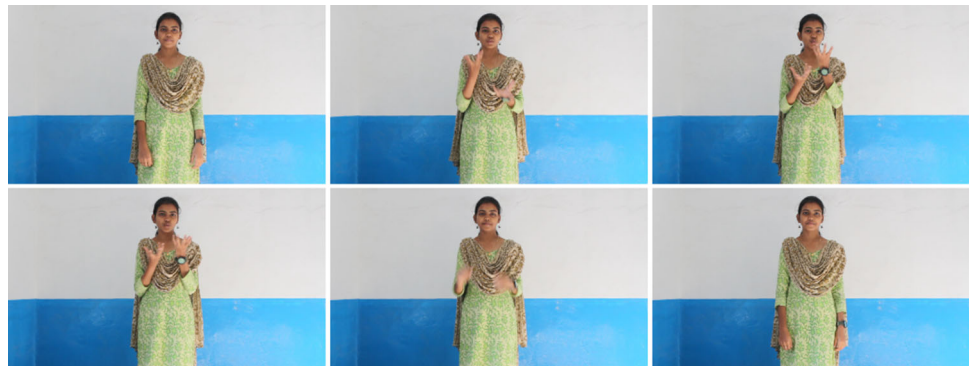
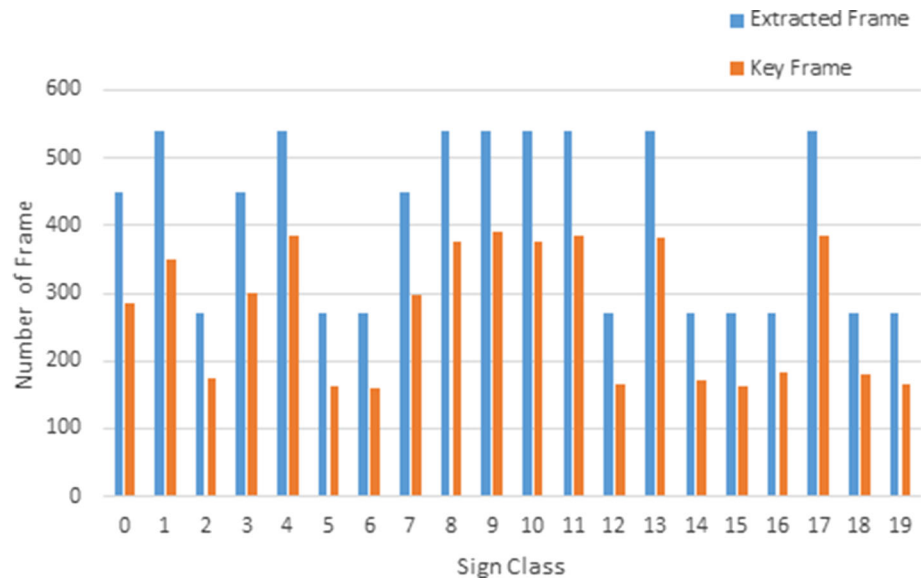
Fig. 2 Extracted frame samples for “Hot” sign**Fig. 3** Extracted frame samples for “Cool” sign



Fig. 4 Example of extracted key-frames from “Student” sign

Fig. 5 Comparative analysis of actual and key-frame numbers



convolutional layer and nonlinear activation layers. The layer structure allows for improved feature extraction and down sampling using max pooling. The down sampling layer is used to improve the anti-distortion capabilities of the network while keeping critical information and minimizing the number of parameters. In this proposed work, the last layer of VGG-19 is excluded, and the network is simply used as a feature extractor. Pre-trained ImageNet weights are used for the proposed work. Using a pre-trained CNN network for feature extraction has improved the efficiency of the proposed HCBSLR model.

The architecture of VGG-19 is similar to the VGG-16 but it has three extra layers of convolution block that also occupies the more memory compared to the VGG-16 network. It has 19 layers, and it takes RGB images of fixed size 224×224 as input data. The feature descriptors are extracted from the last pooling layer. Each feature vector has a size of $m \times n$, where m is the feature vector having size of 4096 and n is the number of frames in sign video. Figure 6 shows the detailed architecture of VGG-19.

3.4 Temporal sequence learning using LSTM model

Traditional deep neural network models have advanced in recent years, but they have many shortcomings, such as the inability to learn sequential data where the output is dependent on past data at each timestamp. Recurrent neural networks (RNNs) address this particular issue. RNNs have a feedback loop that allows them to persist past information. LSTM is a special kind of RNN that is commonly used to tackle problems related to sequential data. LSTM is an advanced RNN which deals with two major shortcomings of RNN: vanishing gradient and exploding gradient. With a chain-like structure made up of recurrent modules, LSTMs can learn extended sequences of information. Each module of LSTM has three gates that control the cell state: input gate, forget gate, and output gate. Cell state is the key component in LSTM module. The cell state runs through the entire module and passes the information through it. The LSTM has the capability to add or remove information in the cell state by using different gate present in the LSTM module [31]. Figure 7 shows the basic structure of LSTM module.

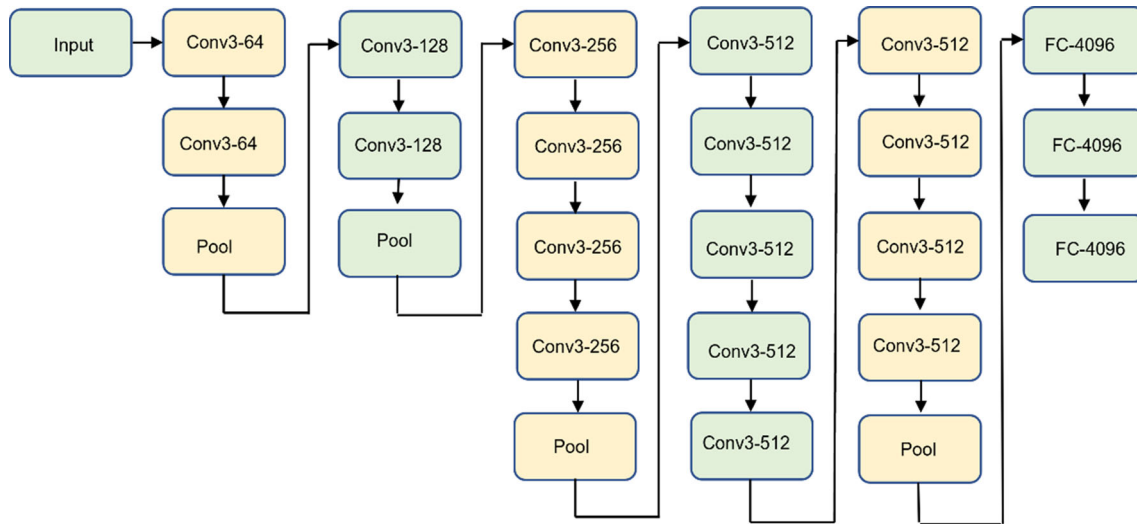


Fig. 6 VGG-19 Architecture

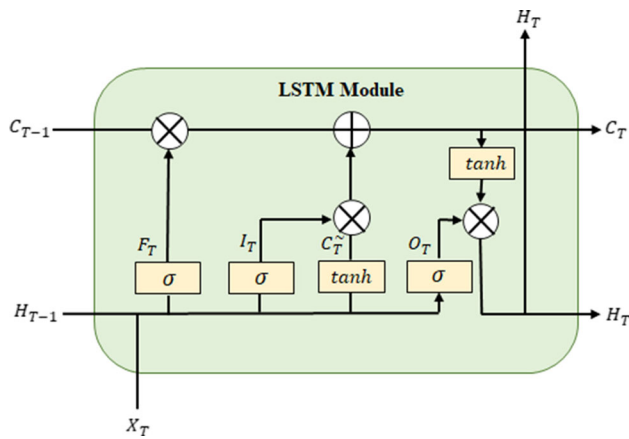


Fig. 7 Architecture of a LSTM module. '×' symbol denotes multiplication and '+' denotes addition. σ and $\tanh$ are the activation functions

The first stage of LSTM is to decide which information from the cell state will be discarded or remembered. This decision is taken by the sigmoid layer of forget gate also called as forget layer. F_T is the output value at forget gate and H_{T-1}, X_T are the inputs.

$$F_T = \sigma(W_F[H_{T-1}, X_T] + b_F) \quad (1)$$

The output values at forget gate are in between 0 and 1. If the value is 1, the cell state remembers and if the value is 0, then cell state completely forgets the value. W_F and b_F are the weight matrix and bias of forget gate, respectively [31].

The input gate stores the information in a new state. The input gate uses a sigmoid layer and a $\tanh$ layer to decide and store information from the new input state X_T . The sigmoid layer present in the input gate decides whether the value will be updated or not. The value is calculated by Eq. (2). If the output is 1, then the value is updated, and if

it is 0, then the value is not updated. W and b are the trainable parameters present in Eq. (2).

$$I_T = \sigma(W_I[H_{T-1}, X_T] + b_I) \quad (2)$$

$\tanh$ function is applied on present input X_T and previous state output H_{T-1} and returns a new candidate value $\tilde{C}_T$. The $\tilde{C}_T$ value is obtained from Eq. (3) at T cell state. W_C represents the weight matrix and b_C is the bias. These are trainable parameters.

$$\tilde{C}_T = \tanh(W_C[H_{T-1}, X_T] + b_C) \quad (3)$$

Now the previous cell state C_{T-1} is updated to new cell state C_T using Eq. (4) based on the value of I_T and $\tilde{C}_T$. While updating to a new state, the previous state value is multiplied by F_T to determine if the previous state's information should be remembered or not.

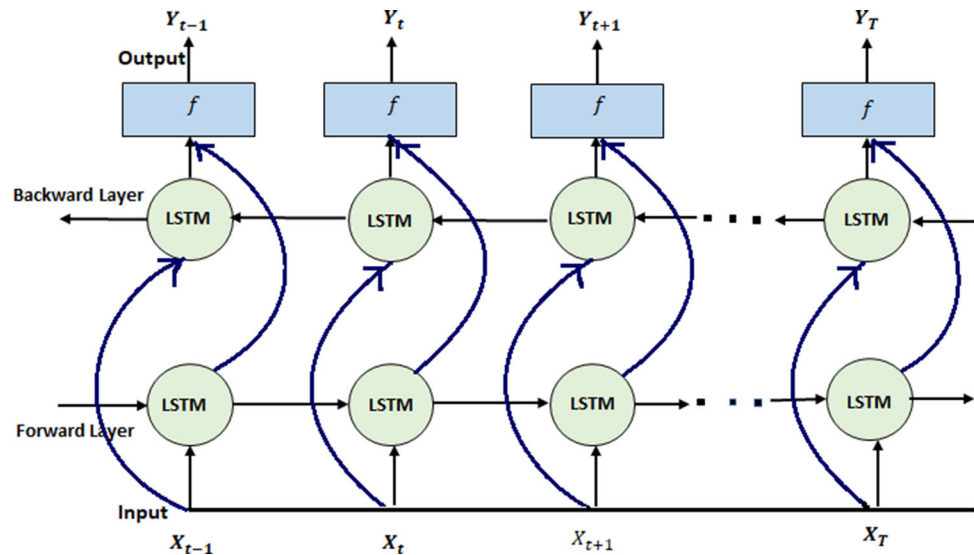
$$C_T = F_T \times C_{T-1} + I_T \times \tilde{C}_T \quad (4)$$

Finally, the following Eqs. (5) and (6) give the output of T^{th} time as well as the cell's final state. The sigmoid function used in the output gate decides which part of the information will be shown as the final output. W_O and b_O are the weight matrix and bias of output gate.

$$O_T = \sigma(W_O[H_{T-1}, X_T] + b_O) \quad (5)$$

$$H_T = O_T \times \tanh(C_T) \quad (6)$$

In this proposed work, BiLSTM is used which is a variant of normal LSTM. BiLSTM model trains with two LSTM network where input sequence is given to the second LSTM network in reverse order. Figure 8 represents the BiLSTM network. BiLSTM helps to get the additional future context through a reverse input sequence, which improves the classification accuracy of the proposed HCBSLR system. In case of dynamic SL recognition, the

Fig. 8 Bidirectional LSTM Architecture

future context helps to recognize the present gesture and it also reduces the ambiguity during classification and improves the accuracy of the system. Thus, BiLSTM is used instead of normal LSTM for ISL word recognition task. The BiLSTM network architecture consists of the input, forward, and backward layers. The number of neurons is same as the feature vector dimension. The BiLSTM layer has 512 hidden units, and each of the 512 hidden units in the BiLSTM layer comprises several memory cells. The fully connected layer consists of an equal number of neurons to the number of sign word classes. A dropout layer is used between two fully connected layers, and the dropout percentage is set as 0.60. Finally, a SoftMax layer is used for classifying sign words.

4 Experimental details

4.1 Implementation details

The proposed model is implemented in Jupiter Notebook using python 3.7 and with the help of Keras and

TensorFlow libraries which are used for DL. Table 2 gives details about different libraries which are used to implement the proposed model. In this proposed work, Adam optimizer is used that helps to reduce the loss during model training. The training parameters include an initial learning rate of 0.0001 and beta1 and beta2 values of 0.99. The proposed model is trained for 1000 epochs. Table 3 summarizes all the parameters used to train the proposed model.

4.2 Testing details

The proposed model has been tested on a 2.40 GHz i5-9300H CPU with 8 GB of RAM on Microsoft Windows 10 platform. A fivefold cross-validation method is used to test the model performance. During feature extraction, different pre-trained networks such as VGG-16, ResNet-50, and InceptionV3 networks are used in this experiment. The accuracy of the proposed model with different pre-trained networks is shown in Table 4.

The performance of the model degrades as the complexity of the network grows since there are not enough

Table 2 Different python libraries used for implementation of proposed model

Python library	Purpose
Keras	For importing CNN model, sequential model, BiLSTM, optimizers
Tensorflow	For importing convolution layer, pooling layer, dropout layer, callback function, etc.
OpenCV	For extracting frames from input videos and for image processing
Glob	For importing input videos from hierarchical directory
Natsorted	For reading frames in sequential order
Sklearn	For confusion matrix, accuracy, dataset split
Numpy	For operating multidimensional array, matrices
Seaborn	For confusion matrix
Os	For directory related operation

video samples for each class in the dataset to train the model, resulting in a loss in performance. The VGG network is a lot simpler compared to ResNet and Inception networks. As a result, the proposed HCBSLR model has given better accuracy with the VGG-19 network.

Table 3 Parameter details used for the proposed model

Parameters	Values
Learning Rate	0.0001
Batch Size	32
Epochs	1000
Weight decay	Le-4
Beta1	0.99
Beta2	0.99

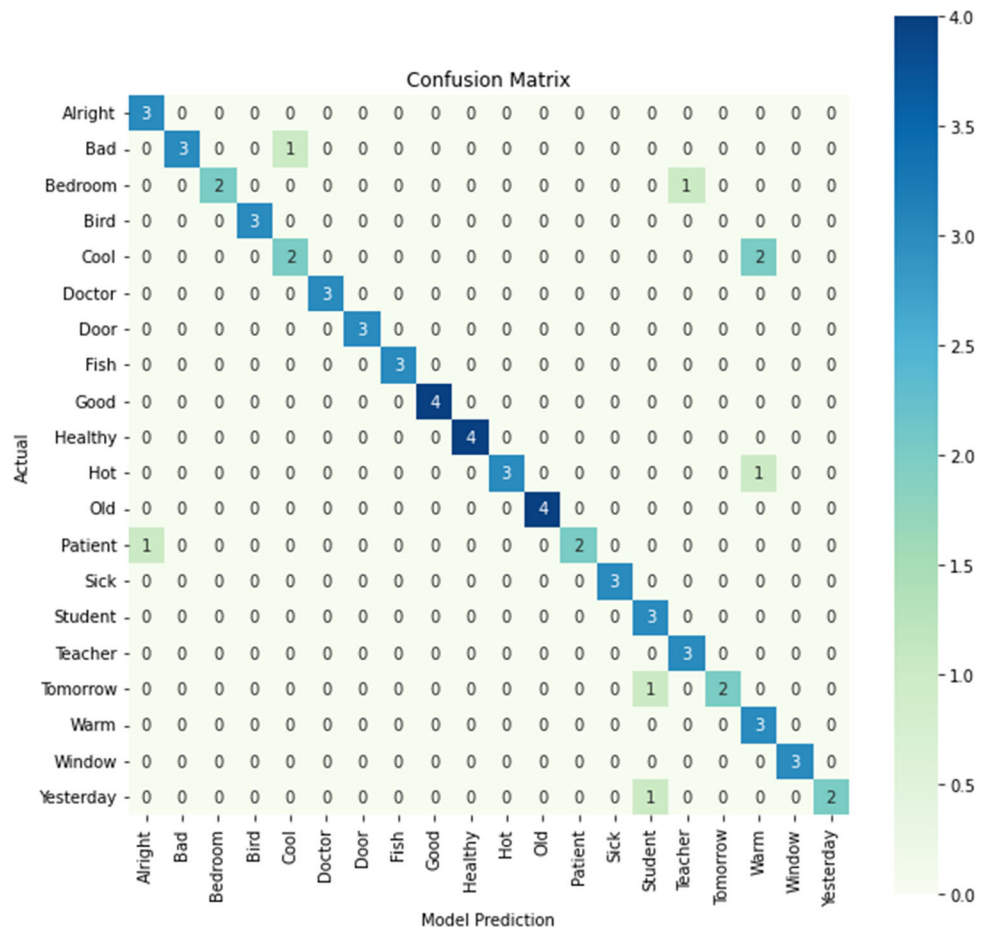
Table 4 Accuracy of the proposed model with different pre-trained network

Model	Accuracy
Proposed model with VGG-16 + Bidirectional LSTM	81.25%
Proposed model with ResNet50 + Bidirectional LSTM	75%
Proposed model with InceptionV3 + Bidirectional LSTM	41%

5 Results and discussion

The performance of the proposed model on ISL words is measured based on classification outcomes and computation time at each step. Figure 9 shows the confusion matrix of the proposed HCBSLR for first fold. From the confusion matrix, it can be observed that most false positive cases are reported for the words “Student” and “Teacher”, whereas the maximum false negative case is reported for the word “Cool”. The reported outcome indicates that maximum misclassified words recognized as either “Student” or “Teacher” on the other side most video samples for the class “Cool” are misclassified into other classes. So, to get the idea about the model’s actual performance, it is required to calculate the precision value and recall value along with the classification accuracy. Recall, also known

Fig. 9 Confusion matrix of the proposed HCBSLR. Different blue shades of the diagonal elements represent the total number of accurately classified samples of ISL words. Non-diagonal elements show the number of misclassified outcomes



as True Positive Rate (TPR) or sensitivity which measures the percentage of actual positively identified samples from total correctly identified samples, Eq. (8) gives the mathematical formula to calculate the recall value. It can be observed that the proposed model got highest recall value for the words “Alright”, “Bad”, “Bird”, “Door”, “Fish”, “Good”, “Healthy”, “Hot”, “Old”, “Sick”, “Student”, “Teacher” and “Warm” whereas got the lowest recall value for word “cool”. The “cool” word is completely different from the other nineteen words as the word represented with finger spelling in the sign video that consist of four gestures: “C”, “O”, “O”, “L” and the proposed model has not learned the sequence properly. Thus, the proposed model is not able to identify the “cool” word in most cases. Table 5 shows the detailed classification statistics of the proposed model. Equations (7), (8) and (9) represent the mathematical formula to measure the value of precision, Recall and F1-Score, respectively.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (7)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (8)$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

Table 5 Classification statistic of proposed model

Class	Precision%	Recall%	F1-Score%
Alright	0.75	1.00	0.86
Bad	1.00	0.90	0.94
Bedroom	1.00	0.67	0.80
Bird	1.00	1.00	1.00
Cool	0.93	0.35	0.48
Doctor	1.00	1.00	1.00
Door	1.00	1.00	1.00
Fish	1.00	1.00	1.00
Good	0.88	1.00	0.93
Healthy	1.00	1.00	1.00
Hot	1.00	0.95	0.97
Old	1.00	1.00	1.00
Patient	1.00	0.67	0.80
Sick	1.00	1.00	1.00
Student	0.60	1.00	0.75
Teacher	0.63	1.00	0.77
Tomorrow	1.00	0.67	0.80
Warm	0.56	1.00	0.71
Window	1.00	0.73	0.84
Yesterday	1.00	0.67	0.80

5.1 Loss

The proposed HCBSLR model uses categorical cross-entropy function that is mostly used for multiclass classification problems. Equation (10) shows mathematical representation of the loss function.

$$\text{Loss} = \sum_{K=1}^m o_K \log o_K^{\wedge} \quad (10)$$

o_K is the actual K th actual output value whereas $o_K^{\wedge}$ is the K th predicted output by the proposed model.

5.2 Accuracy

The accuracy matrix is a popular index for measuring the performance of a classification model. The ratio between the total number of correctly predicted samples and the total number of test samples can be used to determine the accuracy of the proposed model. Equation (11) is the mathematical representation for calculating the accuracy.

$$\text{Accuracy} = \frac{TN + TP}{TN + TP + FN + FP} \quad (11)$$

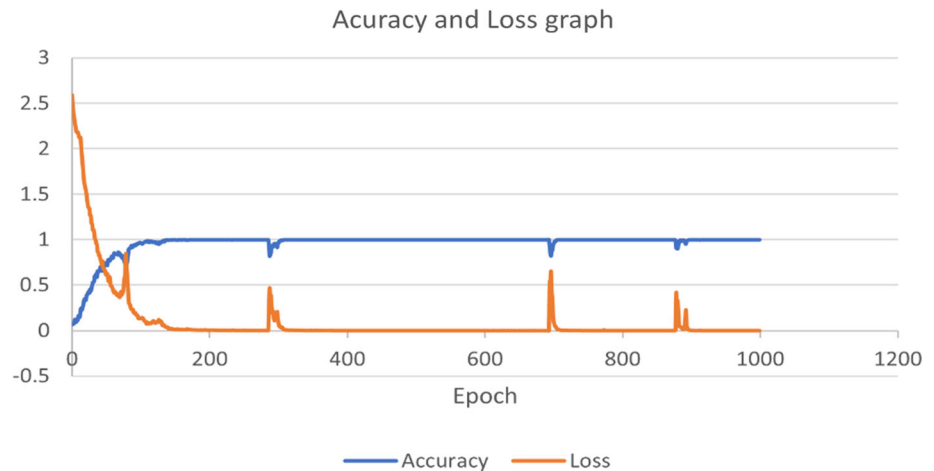
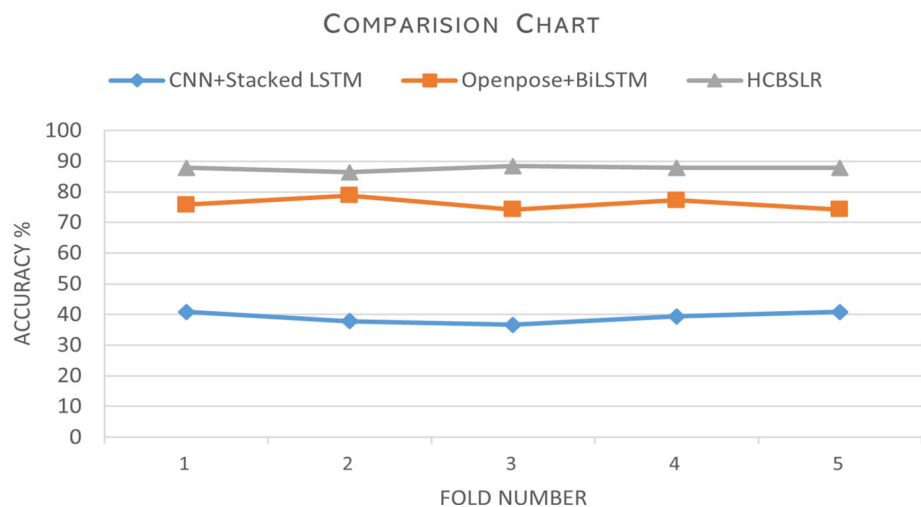
Figure 10 shows the training accuracy and training loss of the proposed model for first fold.

5.3 Comparison with state-of-the-art techniques

ISL recognition is an unknown territory in SL recognition domain. The majority of ISL recognition research has focused on ISL alphabets, numbers and static word recognition task. In last decade, very few works are proposed for ISL word recognition that includes dynamic gestures, and the majority of efforts in this domain are sensor-based. The proposed HCBSLR system is tested using a fivefold cross-validation method, and the system has achieved an average accuracy of 87.67%, which is comparatively better than the existing SLRS [15, 16]. Figure 11 shows the accuracy of different models for each fold. In addition, the feature extraction time and training time of proposed model are less compared to the existing models. Therefore, it can be concluded that the proposed HCBSLR system for ISL word recognition is more accurate and efficient than existing models. The performance of various existing models is compared with the proposed HCBSLR model in the same experimental setup. Table 6 shows a thorough comparison report.

6 Conclusion

This paper reports a vision-based SLRS for ISL word recognition that uses the concept of HD-based key-frame extraction method to eliminate useless or redundant frames,

Fig. 10 Training accuracy and loss graph for proposed model**Fig. 11** Accuracy comparison of different model for each fold**Table 6** Comparative analysis of proposed model with the existing SL recognition model

Author	Method	Accuracy %	Feature extraction time	Training time	Recognition Time
C. Aparna et al. [16]	CNN (Inception V3) + Stacked LSTM	39.14	7 s to 15 s	5 h	2 s to 10 s
A.Sridhar et al. [15]	Open pose + BiLSTM	76.05	10 s to 20 s	6 h	5 s to 12 s
Proposed	HCBSLR	87.67	2 s to 5 s	2 h	1 s to 5 s

which improves the accuracy and efficiency of the system. The proposed HCBSLR system uses VGG-19 as a pre-trained CNN model for spatial feature extraction and a BiLSTM network for temporal feature extraction. The proposed model has shown promising result with better efficiency compared to existing models for ISL word recognition task. The HCBSLR system has achieved an average classification accuracy of 87.67%. Although the proposed model has given satisfactory result, it has certain drawbacks, such as ISL words with similar gesture

confuses the model during recognition which leads to misclassification. This work can be extended to a large vocabulary of ISL words and for sentence recognition. Also, the background of signer is very simple, and in case of SL recognition, background plays an essential role. In future, the proposed model can be tested with different variation in background. However, this work will contribute to the development of SLRS for ISL word recognition with dynamic gesture.

Declarations

Conflict of interest The authors declare that there is no conflict of interest.

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